

Low-Energy Electron Capture in Two-Dimensional Plasmonic Materials

Enabling Conditions, Factor-by-Factor Bounds, and a Mechanism-Agnostic Bounding Experiment

Ryoji Furui

Nano Fusion Design, Hyogo, Japan

www.nanofusion.design

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Abstract

Converting a proton to a neutron by electron capture, $p + e^- \rightarrow n + \nu_e$, costs 0.782 MeV when the neutron is born free — roughly five orders of magnitude beyond any chemical, thermal, or condensed-matter energy scale. Whether any combination of circumstances in a driven solid can enable the process is the question this paper organizes, and the organizing step is to separate the question into its conditions before analyzing any mechanism. Electron capture proceeds only through one of two conjunctions. The field route, with a free final neutron, requires a sustained dressed-electron Lorentz factor $\gamma \geq 2.531$ (equivalently a dimensionless field strength $\xi \geq 2.325$, about $10^{11} \text{ V} \cdot \text{m}^{-1}$ at 4 THz), delivery of the 0.782 MeV at the weak vertex, survival of the material under the enabling field, and phase coherence across the delivery. The binding route — the lattice analog of the solar pep reaction ($p + e^- + p \rightarrow d + \nu_e$) — requires a host nucleus whose neutron separation energy pays the deficit (satisfied here: $e^- + p + {}^{10}\text{B} \rightarrow {}^{11}\text{B} + \nu_e$ is exothermic by 10.672 MeV), Coulomb overlap between the proton and that host, and an electron at the proton. The paper then bounds each factor in isolation, holding every other factor at its most favorable value, in a graphene/borophane platform driven by terahertz acoustic plasmons.

The factor findings. The strongest survivable drive reaches $\xi \approx 0.07$, a factor of 33 below threshold — the threshold field would store the sheet's entire cohesive energy in a nanometer of confinement. Energy delivery at the vertex is suppressed by $10^{4.8}$ – 10^9 orders of magnitude across every regime (multiphoton and Keldysh-quasistatic) and every drive statistics: the natural rescue — treating the driven plasmon as one coherent N-quantum object that delivers all $N \approx 10^8$ quanta at once — cancels exactly, because the coherence bonus is offset by the cost of assembling the N-quantum coupling and the factorially growing energy denominators leave a hyperexponentially small residual; thermal bunching buys the largest statistical bonus available and still loses, and correlated-state squeezing fails three independent checks. Coherence is required for $\sim 10^7$ – 10^8 cycles, while only 1–10 cycles are measured. On the binding route, the zero-point molecular overlap is $\sim 10^{-150 \pm 50}$; the strong channels ${}^{10}\text{B}(p, \alpha){}^7\text{Be}$ and ${}^{11}\text{B}(p, \alpha)2\alpha$ share that overlap and consume any tunneling enhancement twenty-two decades before the weak vertex fires; screening — the one established lever — moves tens of decades against a gap of ~ 120 – 145 ; the electron is within 1 – $10\times$ hydrogenic with sub-percent relativistic corrections; and the proposed supercritical operating environment (hydrogen at 50 MPa and 873 K, a regime for which no graphene data or models of any kind exist) is capped by thermodynamics itself — an isothermal bath cannot be concentrated into coherent field energy, so chaotic feedback cannot escalate to capture.

Reassembled, the channels span ten thousand orders of magnitude of suppression in one commensurable atlas, and the ordering inverts the field's thirty-year emphasis: least suppressed is screened d–d fusion in a

deuterated control (10^{-64} s^{-1} against a 10^{-17} floor), then the overlayer pep analog (10^{-80}), the strong p-B channels (10^{-138}), bound-state capture (10^{-160}) — with the free-neutron field route last by a further two and a half to six powers of ten in the exponent itself. The deliverable is a calibrated experiment: an isotope-swapped ($^{10}\text{B}/^{11}\text{B}$), deuterium-loaded, five-arm, six-readout coincidence protocol, with ambient-neutron capture owned as the irreducible background. A null result bounds the platform’s tunneling-enhancement product; a positive result in the predicted ratios would implicate physics outside every bound in this paper. The claim is deliberately modest: each factor is bounded by theory; what the experiment decides is whether any contested enhancement mechanism operates on this platform at a detectable level.

Authorship and status. A research-stage proposal, derived from the author’s prior work [1]. Portions were prepared with AI assistance under the author’s direction. The author does not hold formal expertise in strong-field QED, nuclear structure, or weak-interaction theory, and the document has not undergone independent expert review. Two disclosures follow from this. First, the central rate results (§5, §8, Appendices B and D) are presented as bounds with stated uncertainties, not as precision calculations; the nuclear-structure inputs (the ^{11}B level scheme and separation energies) are taken from the published $A = 11$ evaluation [2] and AME2020 [3], and the weak-strength distribution over ^{11}B states — which controls the signature branching of §12 — has not been calculated. Second, parts of §7 engage literature that is contested or unreplicated (large metallic screening energies; coherent correlated states); the paper relies on that literature only to set targets for falsification, never as established input, but the author cannot independently adjudicate it. Formal expert review, and the calculations identified in §15, are required before any practical interpretation.

1. The Problem and the Method

The weak process $p + e^- \rightarrow n + \nu_e$ is endothermic by 0.782 MeV when both reactants are at rest and the neutron is born free. No chemical bond, thermal excitation, or condensed-matter degree of freedom comes within five orders of magnitude of closing that gap — the principal reason that claims of nuclear reactions on hydrogen-loaded surfaces have been met with skepticism since 1989 [4]. One route sidesteps the Coulomb barrier of charged-particle fusion by routing the first nuclear step through neutron formation [5]: an ultra-low-momentum (ULM) neutron, once created, is absorbed by a neighboring nucleus with no Coulomb obstruction at all. Everything then hinges on a single question — can the capture step itself proceed at a useful rate? Mainstream analyses of the Widom–Larsen proposal concluded that the required mass renormalization is unattainable and the resulting rates negligible [6, 7]; its authors disputed those critiques [8], invoking collective many-electron correlations that were promised a quantitative treatment which never appeared; further independent critiques reached the same negative verdict by power-balance and field-screening arguments [9, 10, 11], and the one claimed neutron observation was not reproduced [12]. The exchange settled nothing experimentally. This paper reproduces the negative verdict by an independent route — closing, in particular, the coherence loophole on which the rebuttals lean, and extending the closure to non-coherent drive statistics — and then does something the exchange never did: it asks whether the energy deficit itself is channel-dependent.

It is. Section 3 shows that the 0.782 MeV cost applies only to capture into a free neutron; capture into a bound final state — the lattice analog of the solar pep reaction — is strongly exothermic in a boron hydride. That observation does not rescue the proposal: it exchanges an energy gap that nothing can pay for a Coulomb barrier that established physics cannot penetrate at a detectable rate, and it forfeits the neutron route’s defining advantage in the process. But it changes what kind of dead the proposal is — barrier-limited rather than energetically forbidden — and barrier-limited channels have quantified, partially measured

enhancement levers where energy-forbidden ones have none. The paper develops both routes in a single geometry and bounds each honestly — but no longer route by route.

This paper’s structure follows from a simple observation about the problem. In the proposed experimental system, the physical and chemical factors — field amplitude, screening, coherence, Coulomb barrier, nuclear binding, electron density, temperature, pressure — arrive entangled, and any argument conducted at the level of the whole apparatus inherits that entanglement. The paper therefore proceeds in three movements. Section 2 first identifies the specific conditions under which electron capture could occur at all, as two explicit conjunctions of requirements. Sections 3–10 then isolate the contributing factors one at a time: each section holds every other factor at its most favorable value, bounds the factor under study, and closes with a stated finding. Sections 11–15 reassemble the factors into channels, rank every channel in a single commensurable atlas, derive the observable signatures, and specify the experiment that bounds what theory leaves open. Three platform commitments frame the analysis:

- **A two-dimensional geometry.** Graphene [13] supplies the conduction electrons and the hydride borophane [1, 14, 15] supplies the protons; acoustic graphene plasmons (AGPs) confine terahertz fields far more tightly than any three-dimensional plasmon [16]. Critically, neither component need occlude hydrogen in its bulk: the conduction electrons live in the graphene surface and the protons in the adjacent hydride. This all-surface arrangement is consistent with reports that the cold-fusion phenomenon localizes to the surface region even of metals that do not occlude hydrogen at all — W, Pt, and Au [17] — rather than requiring a hydrogen-loaded bulk such as palladium.
- **Two routes, two kinds of failure.** The field route (conditions C1a, C2, C5, C6 of §2) fails by energy delivery — hyperexponential suppression that no statistics lifts (§5). The binding route (C1b, C3, C4) fails by barrier arithmetic — large but lever-rich exponents that measured physics genuinely moves (§8). Keeping the two failure modes separate is half the clarity this separation buys.
- **A calibrated, mechanism-agnostic experiment.** The downstream reaction $n + {}^{10}\text{B} \rightarrow {}^7\text{Li} + {}^4\text{He}$ retains its role as an artifact-resistant triple signature, embedded in a discrimination matrix (isotope swaps, prompt versus delayed 478 keV gammas, alpha spectrometry, activation counting, a deuterated calibration arm) that separates six channel families and treats ambient-neutron capture as the explicit background (§12–§14).

2. The Conditions for Electron Capture

Stripped of platform detail, $p + e^- \rightarrow n + \nu_e$ proceeds in condensed matter only if a specific conjunction of conditions holds, and stating the conditions first is what allows the factors entangled in the apparatus to be examined one at a time. There are two conjunctions, one per possible final state:

- **The field route (free final neutron).** C1a — energetics: the electron must be supplied with 0.782 MeV; as dressed rest mass the threshold is $\xi \geq 2.325$, equivalently a sustained cycle-averaged Lorentz factor $\gamma \geq 2.531$ (§3). C2 — delivery: that energy must arrive at the weak vertex within its duration, in quanta of the drive or by quasistatic tunneling (§5). C5 — survival: the platform must survive the enabling field while a signal integrates (§4, §10). C6 — coherence: any coherent structure invoked in the delivery must persist across it (§6).
- **The binding route (bound final neutron).** C1b — energetics: a host nucleus with neutron separation energy above 0.782 MeV must be adjacent, so the neutron is born bound and the channel is exothermic (§3). C3 — overlap: the proton must reach that nucleus through the intervening Coulomb

barrier (§8). C4 — electron availability: an electron must be present at the proton when it does (§9). Screening (§7) and the environment (§10) act on both conjunctions, with opposite signs.

In the proposed apparatus [1] these conditions arrive entangled: the terahertz drive bears on C1a, C2, C5, and C6 at once; the borophane supplies both the protons and the C1b host nuclei while erecting the C3 barrier; the graphene supplies the C4 electrons and is itself the C5 platform; the supercritical bath touches every material input simultaneously. Table 1 collects the conditions with their quantitative thresholds and the section that isolates each. Two structural facts, established in what follows, organize the conclusions. First, the conditions within each conjunction are mutually antagonistic on this platform: the fields that approach C1a destroy the material that C5 protects; the binding that satisfies C1b erects exactly the barrier that C3 must penetrate; the confinement that strengthens the drive shortens the coherence that C6 demands. Second, the two conjunctions fail in categorically different ways — energy delivery versus barrier arithmetic — so they respond to different levers, and every claimed enhancement in the condensed-matter nuclear literature is an attack on exactly one condition and can be priced accordingly (§11).

Table 1. The enabling conditions for low-energy electron capture, their quantitative thresholds, and where each is isolated and bounded. The field route requires $C1a \wedge C2 \wedge C5 \wedge C6$; the binding route requires $C1b \wedge C3 \wedge C4$.

Condition	Requirement	Isolated in	Status on this platform
C1a	$\xi \geq 2.325$, i.e. $\gamma \geq 2.531$ ($E_{\text{rms}} \approx 1.0 \times 10^{11} \text{ V} \cdot \text{m}^{-1}$ at 4 THz)	§3	factor 30–100 above every measured field and damage ceiling
C1b	host with $S_n > 0.782 \text{ MeV}$ adjacent to the proton	§3	satisfied at zero drive by every host present (Q up to +10.672 MeV)
C2	0.782 MeV delivered at the weak vertex within its duration	§5	suppressed by $10^{4.8} \text{--} 10^9$ orders; regime- and statistics-robust
C3	proton at the host nuclear surface	§8	$P \sim 10^{-150 \pm 50}$; strong channels consume any enhancement 22 decades earlier
C4	electron contact density at the proton	§9	$\sim 1\text{--}10 \times$ hydrogenic; percent-level environmental leverage
C5	platform survives the enabling drive while a signal integrates	§4, §10	ceiling 1–2 decades below C1a; supercritical regime unmeasured, chemically hostile
C6	phase coherence across the delivery	§6	$\sim 10^{7-8}$ cycles required; $\sim 1\text{--}10$ measured

3. The Energy Ledger (C1): Free and Bound Final States

3.1 What binding pays

With evaluated masses [3, 18], the free-space capture Q-value is

$$Q_{\text{free}} = (m_p + m_e - m_n)c^2 = -0.782 \text{ MeV}. \quad (1)$$

This is the number the entire controversy orbits, and it is worth stating exactly what it constrains: capture in which the final neutron carries no binding. If the neutron is instead born bound to an adjacent nucleus — a three-body capture $e^- + p + {}^A\text{Z} \rightarrow {}^{A+1}\text{Z} + \nu_e$ — the ledger gains the daughter’s neutron separation energy,

$$Q_{\text{bound}} = S_n({}^{A+1}\text{Z}) - 0.782 \text{ MeV}, \quad (2)$$

and S_n is measured in MeV for every nucleus in our system. Table 2 evaluates Eq. (2) across the platform (AME2020 atomic masses [3]; all values verify to better than 1 keV). Every host present — boron, carbon, the hydrogen overlayer itself — renders low-energy electron capture exothermic.

Table 2. The bound-final-state ledger. Q-values from AME2020 [3]; the free-space row is the usual premise, true only for a free final neutron. The pep row is established, measured solar physics [19, 20, 21] — the existence proof that three-body electron capture with binding-energy payment is real.

Host	Channel	Q (MeV)	Daughter and fate
(free space)	$p + e^- \rightarrow n + \nu_e$	-0.782	the contested premise; forbidden at low energy
^{10}B	$e^- + p + ^{10}\text{B} \rightarrow ^{11}\text{B} + \nu_e$	+10.672	mostly bound ^{11}B (γ cascade or bare recoil); $^7\text{Li}^* + \alpha$ only from states above 8.66/9.14 MeV (§12)
^{11}B	$e^- + p + ^{11}\text{B} \rightarrow ^{12}\text{B} + \nu_e$	+2.587	^{12}B : β^- , $t_{1/2} = 20.2$ ms, 13.4 MeV endpoint
^{12}C	$e^- + p + ^{12}\text{C} \rightarrow ^{13}\text{C} + \nu_e$	+4.164	stable ^{13}C ; γ cascade
^6Li (dopant)	$e^- + p + ^6\text{Li} \rightarrow ^7\text{Li} + \nu_e$	+6.469	stable ^7Li ; γ cascade
H overlayer	$p + e^- + p \rightarrow d + \nu_e$ (pep)	+1.442	deuterium accumulation; solar pep measured by Borexino [19]

Established physics already exhibits every ingredient of Eq. (2). The solar pep reaction is precisely this process with a second proton as host: its rate is tied to the pp rate through the electron density at the protons [20], and Borexino has measured its neutrinos [19]. Bound-state β decay is the mirror image — binding energy of the final-state electron opening a weak channel — and at GSI it changes the ^{187}Re half-life from 42 Gyr to 33 yr upon full ionization, a factor of 10^9 [22, 23]. Environmental modification of electron-capture rates is likewise measured, and its established scale is instructive: the ^7Be half-life shifts by about one percent between hosts [24]. In degenerate stellar matter, free-neutron capture $e^- + p \rightarrow n + \nu_e$ does proceed — once the electron chemical potential exceeds 1.293 MeV, i.e. at densities near $10^7 \text{ g}\cdot\text{cm}^{-3}$. The ledger, in short, is not exotic; what is exotic is paying it at solid-state densities.

Two further claimed routes deserve one paragraph each, for completeness. Deep-orbit (“deep Dirac level”) proposals [25, 26] assert relativistic electron states bound by ~ 500 keV, which would pay most of the free-channel deficit from atomic physics; mainstream analyses exclude these solutions — they are not square-integrable, and they vanish for any finite nuclear radius [27, 28] — and we do not use them. The Widom–Larsen route [5] is the field route of this paper, developed in §3.2 and bounded in §4–§6.

The dichotomy that organizes everything that follows: the field route keeps the free final state and pays Q_{free} from the drive — it needs a threshold field and fails by energy delivery (§5). The binding route takes $Q_{\text{bound}} > 0$ and needs no drive at all — but the proton must reach the host nucleus, so it fails by Coulomb-barrier penetration (§8), and it surrenders the “no Coulomb obstruction” advantage that motivated the neutron route in the first place [5]. Stating that surrender plainly is part of the ledger: the binding route is pursued here as honest bookkeeping of an otherwise-uncovered loophole, and because it — not the field route — turns out to set the experiment’s live targets.

3.2 The field-route threshold: dressed mass, relativistically read

A single conduction electron cannot pay the deficit kinetically: graphene's carriers are Dirac quasiparticles with Fermi energies of 0.1–0.5 eV — six to seven orders of magnitude short. (The often-quoted $\frac{1}{2}m_e v_F^2 \approx 2.84$ eV presumes a free electron of vacuum mass at the Fermi velocity; either way the conclusion is unchanged.) The field route relies instead on the intensity-dependent (Volkov) rest mass that a charge acquires in an intense oscillating field [29, 30]:

$$\tilde{m}c^2 = m_e c^2 \sqrt{1+\xi^2}, \quad \xi \equiv eA_{\text{rms}}/(m_e c). \quad (3)$$

Capture becomes energetically allowed once the dressed mass reaches the neutron–proton splitting, $\tilde{m} \geq m_n - m_p$; equivalently $\xi \geq \xi_{\text{th}} = \sqrt{((m_n - m_p)/m_e)^2 - 1} \approx 2.325$ (Appendix A). Figure 1 shows the crossing.

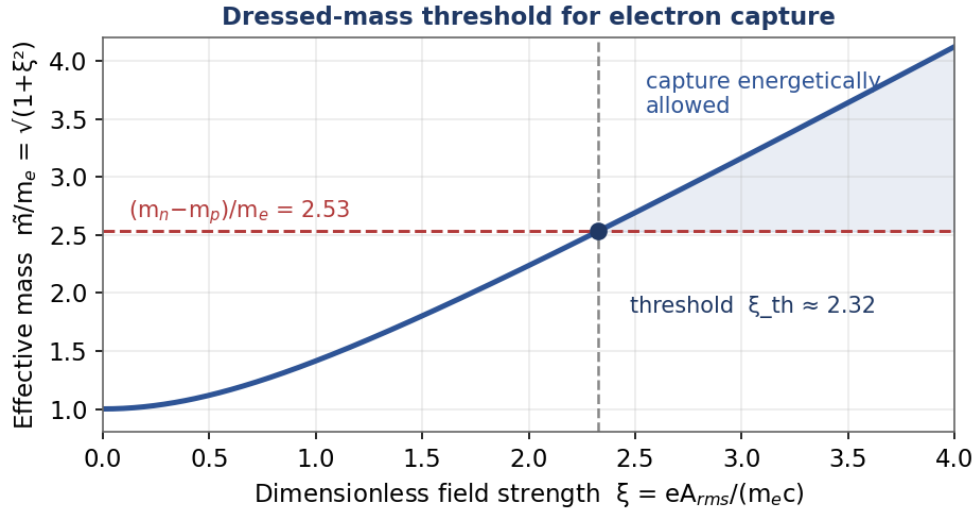


Figure 1. The effective rest mass grows with the dimensionless field strength ξ . Capture becomes energetically allowed once $\tilde{m}/m_e$ exceeds $(m_n - m_p)/m_e = 2.531$, which occurs at $\xi_{\text{th}} \approx 2.325$.

At 4 THz this threshold corresponds to $E_{\text{rms}} \approx 1.0 \times 10^{11} \text{ V} \cdot \text{m}^{-1}$ — one to two decades above the strongest sustained near-fields ever reported in graphene, and above the material's measured damage thresholds (§4). For calibration, the SLAC E144 experiment operated at $\xi \lesssim 0.4$ [31], well below this regime; it confirmed dressed-electron kinematics but not the conversion of field energy into rest mass for a weak decay.

Two caveats attach to the threshold itself. First, Eq. (3) is derived for a propagating plane wave — a null field — and E144 calibrates only that case. A plasmon near-field is longitudinal and evanescent, with phase velocity $\omega/q \approx c/120$ here; no theorem transplants the Volkov dispersion to such a mode. The quiver energy is real, but whether it enters the weak-vertex kinematics as rest mass is an assumption of this paper, not a result. Second, the carriers being dressed are Dirac quasiparticles of the graphene band structure; the dressing argument treats the physical electron, of vacuum mass m_e , and uses the band structure only to set the achievable field. Both caveats are inherited by everything that follows: $\xi \geq \xi_{\text{th}}$ should be read as conjectural for this geometry — which only strengthens the conclusion of §5, since even the conjecture, granted in full, does not survive the rate analysis.

A relativistic reading makes the threshold's meaning transparent, and answers the natural question about electron–proton kinematics. Since $\tilde{m}/m_e = \sqrt{1+\xi^2}$ is the cycle-averaged Lorentz factor of the driven electron, $\xi \geq \xi_{\text{th}}$ is nothing more or less than the demand that the electron reach $\gamma \geq (m_n - m_p)/m_e = 2.531$, i.e.

$v \geq 0.92c$. The dressed-mass threshold and the relativistic-electron threshold are the same statement: free-channel capture opens for any electron whose total energy reaches 1.293 MeV — which is why the process runs freely in neutron-star crusts, where the electron chemical potential exceeds that value, and would run in an ordinary accelerator beam at 0.8 MeV kinetic energy. What this lattice can supply falls short by a quantified margin: at the survivable field ceiling the quiver energy of a conduction electron is $(\gamma-1)m_e c^2 \approx (\xi^2/2)m_e c^2 \approx 1.3 \text{ keV}$ ($\xi \approx 0.07$), six hundred times below threshold; and the kinetic “plasmon rebound” picture of the concept paper [1] — carriers at the Fermi velocity pushing electrons onto protons — is bounded by the same ledger at the eV scale, five orders deeper still. Graphene’s celebrated pseudo-relativistic band structure does not help: its carriers are “relativistic” only with respect to the effective light speed $v_F \approx c/300$, and the weak vertex cares about real Lorentz factors in vacuum.

Finding (C1). The energetic condition is channel-dependent. C1a requires $\gamma \geq 2.531$ — a field of $1.0 \times 10^{11} \text{ V}\cdot\text{m}^{-1}$ at 4 THz, thirty-fold above the survivable ceiling of §4 — while C1b is satisfied at zero drive by every host nucleus in the platform, with Q up to +10.672 MeV. The energy-gap objection, as usually stated, binds only the free-neutron channel.

4. Factor I — The Drive: Amplitude, Confinement, Survival (C5)

The proton does not sit in the bare mode field but in a screened, geometrically reduced local field. The geometric part is benign: writing $A_{\text{loc}} = \eta_{\text{loc}} A_0$, the local-field factor is a product of evanescent decay, a bound-state form factor, and a Debye–Waller factor (Appendix C), and at $q \approx 10^7 \text{ m}^{-1}$ with an adsorbate height of $\sim 1 \text{ \AA}$ the product is $\eta_{\text{loc}} \approx 0.999$ — the two-dimensional geometry removes the geometric suppression almost entirely. The screening part is not benign, and because it acts with opposite signs on the two routes it is treated as its own factor in §7; this section isolates what the drive can deliver and what the platform survives.

Dispersion and confinement. Intrinsic graphene plasmons disperse as $\omega \propto \sqrt{q}$ [32]; in the gated geometry relevant here, the acoustic branch is instead nearly linear, $\omega \approx v_A q$ with v_A a few times the Fermi velocity [16, 33]. Either branch gives, at 4 THz, a wavevector $q \approx 10^7 \text{ m}^{-1}$ — about 100 times the free-space value, with measured vertical confinement reaching $\lambda_0/26,000$ (0.3 nm) in acoustic cavities [34] and single-cavity mode volumes near 10^{-9} of the free-space diffraction limit [35]. A 2026 demonstration of AGP Fabry–Pérot cavities at 1.8–2.5 THz in scalable CVD graphene supports the proposed geometry directly [36].

Field amplitude — stated honestly by category. Far-field THz amplitudes reach $\sim 10^7\text{--}10^9 \text{ V}\cdot\text{m}^{-1}$; nanogap and tip-enhanced near-fields reach $10^9\text{--}10^{10}$; the highest reported values, $10^{10}\text{--}10^{11}$, are sub-nanometer tip-peak or single-cycle figures, not coherent fields filling a mode volume [37, 38, 39]. No experiment reports a sustained plasmonic field at $10^{10} \text{ V}\cdot\text{m}^{-1}$ or above. Two independent arguments pin the sustainable ceiling near $10^{9.5} \text{ V}\cdot\text{m}^{-1}$: measured damage — multi-pulse graphene ablation begins at peak fields of $5\text{--}8 \times 10^8 \text{ V}\cdot\text{m}^{-1}$ and single-shot at $\approx 3 \times 10^9$, with hBN gap breakdown at $0.3\text{--}2 \times 10^9$ — and mode energetics: the single-quantum field of the record acoustic cavity is $\approx 5 \times 10^7 \text{ V}\cdot\text{m}^{-1}$, so $10^9 \text{ V}\cdot\text{m}^{-1}$ corresponds to a comfortable ~ 500 quanta, while the 10^{11} threshold field would demand $\sim 5 \times 10^6$ quanta, an energy density of $1.8 \times 10^{11} \text{ J}\cdot\text{m}^{-3}$ — matching or exceeding graphene’s entire cohesive energy density (see below).

Energetics makes the survival constraint explicit: at $10^{11} \text{ V}\cdot\text{m}^{-1}$ the field energy density is $\frac{1}{2}\epsilon_0 E^2 \approx 4 \times 10^{10} \text{ J}\cdot\text{m}^{-3}$, so even one nanometer of confinement stores $\approx 44 \text{ J}\cdot\text{m}^{-2}$ — essentially graphene’s entire cohesive energy ($\approx 45 \text{ J}\cdot\text{m}^{-2}$ at $\sim 7.4 \text{ eV}$ per atom). The lattice does not survive the field-route threshold field;

sustained drives stop one to two decades short. Material survival is therefore a constraint of the proposal, not a loose end.

The same ceiling prices every “effective energy” the drive can give the proton — the quantity the binding route’s enhancement claims will need (§7). The ponderomotive (quiver) energy of a proton in the plasmon field, $U_p = e^2 E_{\text{rms}}^2 / (2m_p \omega^2)$, is ~ 2 meV at $10^8 \text{ V} \cdot \text{m}^{-1}$ and 2.4 THz, ~ 0.2 eV at 10^9 , and ~ 21 eV even at the already-un survivable 10^{10} . Kilo-electron-volt effective collision energies — the scale at which beam-measured screening would make p–B tunneling exponents merely large instead of astronomical — are not available from this drive at any survivable amplitude. Any analysis that quotes a keV-scale effective energy for this platform has assumed its conclusion.

Finding (C5, drive side). Two independent ceilings — measured damage and mode energetics — cap the sustained drive near $3 \times 10^9 \text{ V} \cdot \text{m}^{-1}$, i.e. $\xi \approx 0.07$, a factor of 33 below C1a; the threshold field would store the sheet’s entire cohesive energy in a nanometer of confinement, and holding it in a single mode would require dissipation eight orders beyond any cooling (Appendix E). Conditions C1a and C5 cannot hold simultaneously on this platform, and the proton quiver budget (2 meV–21 eV) forecloses every keV-scale “effective energy” scenario at survivable amplitudes.

5. Factor II — Delivery at the Weak Vertex (C2)

The dressed-mass threshold is an energetic statement, not a rate. The energy must be supplied to the weak vertex in quanta of the driving mode, and this section asks whether plasmon coherence — or any other drive statistics — can make that affordable. The answer is no, and the argument comes in three parts that are easy to conflate: §5.2 gives the dynamical cancellation that defeats the coherent rescue, §5.4 extends the bound from coherent to thermal and squeezed drives, and §5.5 shows the conclusion is not an artifact of the multiphoton limit.

5.1 The multiphoton barrier

Because energy arrives in quanta of the drive, the process is intrinsically of very high order,

$$N = E_C / \hbar \omega, \quad E_C = 782 \text{ keV}. \quad (4)$$

At a representative AGP energy $\hbar \omega \approx 10$ meV (≈ 2.4 THz), this is $N \approx 7.8 \times 10^7$. A coherent (classical) field gives the standard Volkov-type amplitude for absorbing N quanta; in the regime $N \gg \xi$ relevant here its square obeys the Bessel-function asymptotic

$$|J_N(\xi)|^2 \approx (2\pi N)^{-1} (e\xi/2N)^{2N}, \quad (5)$$

hyperexponentially small. (The full strong-field amplitude is a generalized Bessel function of two arguments; Eq. (5) is the controlling asymptotic and is what Table 3 evaluates.) This is the barrier that any rescue must overcome.

5.2 The coherent N-quantum rescue, and why it cancels

The natural rescue treats the AGP as a coherent N-quantum object with an effective vertex,

$$H_{\text{int}} = g_{\text{eff}} [(a^\dagger)^N \sigma_- + a^N \sigma_+], \quad \Gamma \propto |g_{\text{eff}}|^2 \langle n^N \rangle \approx |g_{\text{eff}}|^2 n^N, \quad (6)$$

from which the required occupation appears to fall far below the process order N — to experimentally accessible values — once n^N is large. The error is that g_{eff} is not a free constant. The operator $(a^\dagger)^N$ is generated only at N -th order in the fundamental single-quantum coupling g_0 , through $N-1$ energy-

denominator factors — and the bookkeeping of those denominators matters. The k -th intermediate state is detuned by k quanta, $\Delta_k \approx k\hbar\omega$, so the product of denominators grows factorially, not geometrically:

$$g_{\text{eff}} \sim g_0^N / [(N-1)!(\hbar\omega)^{N-1}] \Rightarrow \Gamma \propto (g_0\sqrt{n}/\hbar\omega)^{2N} / (N!)^2 = \xi^{2N} / (N!)^2 \approx (e\xi/2N)^{2N}, \quad (7)$$

by Stirling — precisely the Bessel asymptotic of Eq. (5). (A constant denominator $\Delta \sim \hbar\omega_0$ would instead give $\Gamma \propto \xi^{2N}$ alone, which at $\xi = 2.3$ is an astronomical enhancement, not a suppression; the factorial growth of the denominators is the actual source of the multiphoton barrier, derived in Appendix B.) In words: coherence does multiply the rate by the bonus n^N , and assembling the N -quantum vertex out of the one-quantum coupling costs exactly g_0^N ; those two factors combine into ξ^{2N} — and the $(N!)^2$ from the intermediate-state denominators then supplies the hyperexponential suppression. Keeping the bonus while treating g_{eff} as order-unity is exactly what produces the false conclusion; restoring the physical coupling returns $|J_N(\xi)|^2$ (Figure 2). Here $\xi = g_0\sqrt{n}/\hbar\omega$ is the quantum-Rabi combination that corresponds, for a macroscopically occupied mode, to the classical strong-field parameter of Eq. (3) — a correspondence, not a new definition. A macroscopically occupied plasmon is a coherent state, so this picture does not sit outside the multiphoton result — it reproduces it.

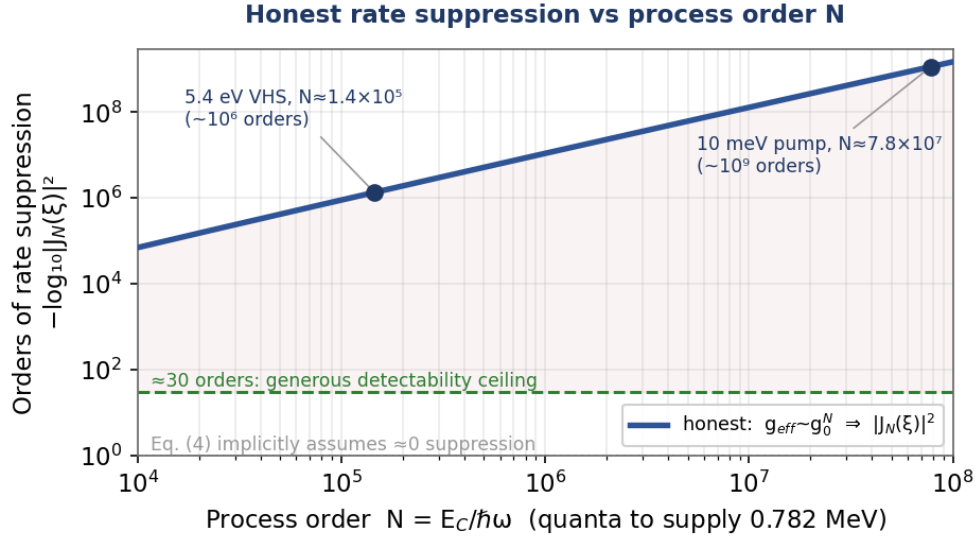


Figure 2. Honest rate suppression $-\log_{10}|J_N(\xi)|^2$ versus process order N , at $\xi = 2.3$. Even the least-suppressed hypothetical drive considered (a 5.4 eV Van Hove resonance, $N \approx 1.4 \times 10^5$) lies roughly 10^6 orders below any plausible detectability ceiling. An order-independent coupling, Eq. (6), would place this entire curve at zero. The quasistatic channel (§5.5) is separately fatal; Table 3 lists both.

5.3 The kinematic floor

A second, purely kinematic obstruction independently invalidates the rescue formula — though it does not by itself bound the rate. Because $a^N|n\rangle = \sqrt{[n!/(n-N)!]}|n-N\rangle$ vanishes for $n < N$, the mode must contain at least N quanta before it can supply N . Any expression that returns a required occupation below N is therefore unphysical. For the THz drive the rescue formula does exactly that, giving $n^* \approx 4.3 \times 10^6$ against $N = 7.8 \times 10^7$ (Figure 3) — a factor of 18 into the forbidden region; the 0.20 eV drive of Table 3 behaves the same way. (For the 5.4 eV drive the formula's $n^* \approx 1.2 \times 10^6$ exceeds its $N \approx 1.4 \times 10^5$, so the floor is met there — which is why the floor alone settles nothing: the magnitude of the suppression rests entirely on the

dynamical argument of §5.2, which applies to every row.) Enforcing $n \geq N$ removes the apparent advantage; the rate still carries the $|J_N|^2$ suppression.

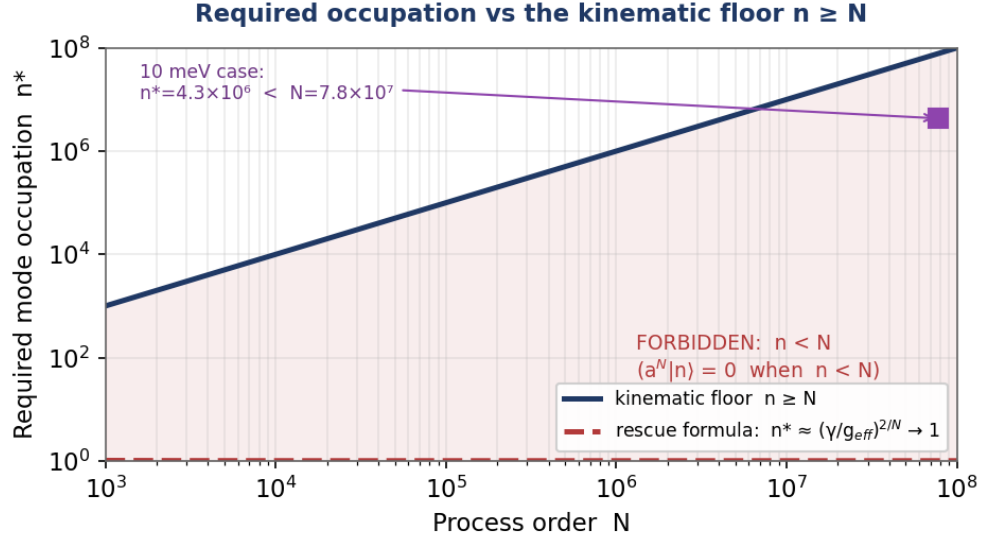


Figure 3. For the THz drive, the coherent-rescue formula drives the required occupation n^* below the process order N , into the region forbidden by $a^N|n\rangle = 0$ for $n < N$. The physical floor is $n \geq N$. (For the 5.4 eV Van Hove drive the formula satisfies the floor; the rate suppression of §5.2 applies regardless.)

5.4 Beyond coherent states: thermal and squeezed drives

It is tempting to assume the drive is coherent “or less ordered” and stop there. That leaves a loophole, because non-classical statistics genuinely enhance N -quantum absorption: the normally-ordered moment of a thermal state is $\langle n^N \rangle = N! \cdot \bar{n}^N$ — a factorial bonus over the coherent $\bar{n}^N$, and the largest available to any phase-insensitive Gaussian state. It is not enough:

$$\Gamma_{\text{th}} \propto N! \cdot (e\xi/2N)^{2N} \cdot \bar{n}^N / \bar{n}^N \approx (e\xi^2/4N)^N, \quad (8)$$

which remains hyperexponentially small whenever $N > e\xi^2/4 \approx 3.7$ — i.e. for every drive in Table 3. The factorial cancels almost exactly half of the log-suppression (at the THz drive, from -1.2×10^9 to -5.7×10^8) and gets no further. The published mechanism that claims to go further is the coherent-correlated-state (CCS) program of Vysotskii and Vysotsky [40, 41]: hyper-squeezed states of a parametrically driven oscillator whose energy fluctuations grow as $G = 1/\sqrt{1-r^2}$ with the Schrödinger–Robertson correlation coefficient r — the cross-correlation term in the position–momentum uncertainty relation — driven toward 1. Three independent obstructions close that door here, each on the program’s own terms. Magnitude: delivering 782 keV from 10 meV quanta requires fluctuation growth $G \approx 8 \times 10^7$, i.e. $1-|r| \approx 8 \times 10^{-17}$; the largest values entertained in the CCS literature are $G \sim 10^3$ – 10^4 [40, 42]. Formation: CCS forms by parametric modulation near $\Omega = 2\omega_0$ over 50–1500 oscillator periods — tens of picoseconds for terahertz modes — while measured plasmon coherence times are 0.1–1 ps (§6), and an independent analysis finds the required squeezing collisionally fragile at exactly this point [43]. Principle: fluctuation-borrowing of the CCS type addresses barrier penetration, where the borrowed energy is returned; electron capture must deliver real energy to an escaping neutrino, so the borrowing window is bounded by the weak-vertex duration, not the tunneling time. The bound’s assumption about the drive statistics (assumption (iii), §5.6)

is thus justified rather than merely posited: coherent, thermal, and squeezed drives are all covered, the first two exactly and the third by the three obstructions above.

5.5 The quasistatic channel and the Keldysh crossover

The multiphoton suppression of §5.1 is one limit of a more general competition, governed by the Keldysh parameter [44], which measures where a process sits between the multiphoton and tunneling limits,

$$\gamma_K = \omega\sqrt{(2m_e E_C)/(eE)} = \sqrt{(2E_C/m_e c^2)/\xi} \approx 1.75/\xi, \quad (9)$$

which is independent of drive frequency at fixed ξ and equals 0.75 at threshold. Near and above ξ_{th} the process therefore sits in the quasistatic (tunneling) regime, where the suppression is governed not by $|J_N(\xi)|^2$ but by a Zener-type exponent,

$$\Gamma \propto \exp[-(4/3)\sqrt{(2m_e)E_C^{3/2}/(eE\hbar)}], \quad (10)$$

up to $O(1)$ factors in the exponent (relativistic corrections enter at the same order, since $E_C \approx 1.5 m_e c^2$). This channel needs no long-lived phase coherence — its formation time is femtoseconds — and it is the least suppressed available. It is still fatal, at every field the system can produce or even imagine. At the strongest survivable sustained drive ($\approx 3 \times 10^9 \text{ V} \cdot \text{m}^{-1}$, §4) the exponent is $\approx -7 \times 10^8$ in $\log_{10}$; at $10^{10} \text{ V} \cdot \text{m}^{-1}$, already above the measured damage thresholds, it is $\approx -2 \times 10^8$; at the hypothetical 10 meV threshold field ($6 \times 10^{10} \text{ V} \cdot \text{m}^{-1}$) it is $\approx -3.4 \times 10^7$; and even at the 5.4 eV Van Hove drive's threshold field ($3.3 \times 10^{13} \text{ V} \cdot \text{m}^{-1}$ — three hundred times the field that already deposits graphene's cohesive energy, §4) it is $\approx -6.3 \times 10^4$. Table 3 lists both regimes side by side. Quoting the weakest suppression as the operative bound: the least-suppressed case considered still falls tens of thousands of orders below any detectability ceiling, and every physically survivable case falls short by hundreds of millions. The bound is regime-robust: no multiphoton-to-tunneling crossover rescues it.

5.6 The bound

Evaluating the honest rate at $\xi \approx 2.3$ across the drives considered gives Table 3. A robustness row is included for the Widom–Larsen field itself: granting their contested estimate of the proton-oscillation drive ($\hbar\Omega \approx 60 \text{ meV}$, $\xi \approx 18.7$, $E \approx 2.9 \times 10^{12} \text{ V} \cdot \text{m}^{-1}$ [5, 8]) — every element of which the critiques dispute [6, 7, 9, 10, 11] — the multiphoton suppression still has a multiphoton $\log_{10} \approx -1.5 \times 10^8$ and a quasistatic $\log_{10} \approx -7 \times 10^5$. The bound does not depend on who is right about the achievable field.

Table 3. The field route's rate bound. Only the single-quantum channel ($N = 1$, a 0.782 MeV γ -ray) is unsuppressed — and no plasmon or laser exists at that energy. Columns three to five are $\log_{10}$ of the suppression factor multiplying the weak-vertex prefactor ($\sim 10^{-7} \text{ s}^{-1}$ per proton, Appendix D); the thermal column grants the maximal phase-insensitive Gaussian bonus $N!$ (§5.4). Every at-threshold entry requires a field at or beyond the material-survival ceiling of §4 — the rows are hypothetical by construction, which makes the bound conservative twice over. The final column gives the coherent-rescue formula's required occupation and its status against the $n \geq N$ floor (§5.3).

Coherent drive $\hbar\omega$	$N = E_C/\hbar\omega$	Multiphoton $\log_{10} J_N ^2$	Thermal drive ($\times N!$)	Quasistatic $\log_{10} \Gamma$	Rescue formula n^*
10 meV (THz plasmon)	7.8×10^7	$\approx -1.2 \times 10^9$	$\approx -5.7 \times 10^8$	$\approx -3.4 \times 10^7$	$4.3 \times 10^6 < N$: unphysical
60 meV (Widom–Larsen drive, $\xi = 18.7$)	1.3×10^7	$\approx -1.5 \times 10^8$	$\approx -7 \times 10^7$	$\approx -7 \times 10^5$	—

0.20 eV (optical phonon)	3.9×10^6	$\approx -5 \times 10^7$	$\approx -2.4 \times 10^7$	$\approx -2 \times 10^6$	$< N$: unphysical
5.4 eV (M-point Van Hove)	1.4×10^5	$\approx -1.4 \times 10^6$	$\approx -6.7 \times 10^5$	$\approx -6.3 \times 10^4$	$1.2 \times 10^6 > N$: floor met; rate still suppressed
0.782 MeV (γ , $N = 1$)	1	O(1), unsuppressed	O(1)	O(1)	trivial

Finding (C2). None of the available levers crosses zero. Confinement raises g_0 per quantum, but it enters the rate only through ξ , and pushing ξ to order unity is nothing other than the $\sim 10^{11} \text{ V} \cdot \text{m}^{-1}$ dressed-mass field by another name. The capture rate is bounded at least $\sim 6 \times 10^4$ orders of magnitude below detection for the least-suppressed hypothetical drive, $\sim 2 \times 10^8$ orders for any drive the lattice survives, and $\sim 3 \times 10^7$ for any terahertz drive. The bound rests on three explicit assumptions: (i) the weak vertex acts on a single electron; (ii) the drive couples bilinearly to one bosonic mode; (iii) the mode is in a coherent, thermal, or squeezed state — the last covered explicitly (§5.4). Decoherence is not an assumption but a worsening: the bound is derived in the fully coherent best case, and any decoherence only lowers the rate further (§6). One developed framework exists that drops assumptions (i) and (iii) — Hagelstein’s lossy spin-boson program [45], in which many Dicke-coupled two-level systems exchange $N \sim 10^8$ quanta with a highly excited oscillator and state-selective loss removes the destructive interference. Even granting its strong-coupling onset condition, the weak vertex here ($U \sim G_F |\psi_e(0)|^2 \sim 10^{-13} \text{ eV}$) misses that onset by at least nine orders of magnitude in the coupling combination $N \cdot U \cdot \sqrt{n_0}$ — it would require $\sim 10^{30}$ plasmon quanta — and a later correction by the same author concedes that generic off-resonant loss, which is what plasmon damping supplies, does not produce the effect. Within assumptions (i)–(iii) the threshold is simply not lowered; a non-negligible field-route rate would require a genuinely new, strongly correlated many-body channel that is not established. We therefore treat the rate as an upper bound (Appendix B) and design the test of §14 so that it does not depend on it.

6. Factor III — Coherence (C6)

Measured plasmon lifetimes are 0.1–1 ps [37, 38, 46], giving a single-cycle coherence factor $f_{\text{coh}} = \omega\tau/(1+\omega\tau)$ of 0.60–0.94 at the representative $\hbar\omega \approx 10 \text{ meV}$ — close to unity over one cycle, with a theoretical ceiling near 3 ps on hBN. But the N -quantum amplitude of §5 requires the drive to stay phase-coherent across all N absorption events — over $N \approx 8 \times 10^7$ cycles, about 30 μs , seven to eight orders of magnitude beyond any measured plasmon coherence time. The logical role of this fact deserves care: the §5 bound is computed in the fully coherent best case and does not rest on decoherence; decoherence is the reason reality falls below even that bound. The same wall closes the CCS formation route of §5.4 independently of its magnitude problem. The quasistatic channel of §5.5 evades the coherence requirement — its formation time is femtoseconds — but is excluded by its own exponent; the failure modes are complementary.

Finding (C6). Delivery at order N requires phase coherence across $\sim 10^7$ – 10^8 cycles; measured plasmon coherence spans one to ten cycles. The §5 bound is computed in the fully coherent best case, so C6’s failure subtracts nothing from it — it is the reason reality falls below even that bound — and the same wall independently forecloses correlated-state formation (§5.4). The binding route, a zero-point process, is indifferent to drive decoherence: one more sense in which it is the better-posed channel.

7. Factor IV — Screening: One Mechanism, Two Opposite Roles

Two different objections — and two different hopes — share the name “screening,” and keeping them apart is essential. Screening of the *drive* reduces the field the proton feels and pushes C1a further away. Screening of the *internuclear barrier* shortens the Coulomb wall between proton and host and is the one established lever on C3. The free-neutron energy deficit itself has no Coulomb barrier, so no amount of screening touches the C1a ledger; conflating the two roles has confused this literature for years.

The drive side. Thomas–Fermi screening of a longitudinal field at $q \approx 10^7 \text{ m}^{-1}$ in doped graphene gives $\epsilon(q) \approx 1 + q_{\text{TF}}/q$, which evaluates to $\epsilon \approx 50\text{--}110$ for hBN-encapsulated samples ($\kappa \approx 4\text{--}5$, $E_{\text{F}} \approx 0.2\text{--}0.4 \text{ eV}$) and to $\epsilon \approx 250\text{--}500$ for suspended graphene at the same doping (Appendix C) — one of the strongest penalties in the system. The strong-field, many-body screening of an intense longitudinal field at a charged adsorbate remains an open calculation (§15). The geometric local-field factor of §4 is necessary, but nowhere near sufficient, to retire this side of the objection.

Because the binding route’s gap is barrier arithmetic, the levers that act on it are measurable, and several have been measured. They form a ladder, and the paper’s rigor depends on labeling each rung. Established, uncontested: electron screening in gas-target beam experiments — $U_{\text{e}} = 25 \pm 5 \text{ eV}$ for $d(d,p)t$ in D_2 , in agreement with the adiabatic limit [47]. Established in direction, contested in magnitude: screening of keV deuteron beams in metals, reported at 100–800 eV across three independent groups [48, 49] but exceeding adiabatic theory by factors of 2–6, with a careful reanalysis attributing much of the excess to deuteron-density and stopping-power systematics [50] and modern measurements converging toward $\sim 105\text{--}130 \text{ eV}$ [51]; for protons on boron the adiabatic bound is $\approx 350 \text{ eV}$, and the one atomic-target measurement ($430 \pm 80 \text{ eV}$ [49]) is consistent with it. Established extreme anchor: muon catalysis — a 207-fold heavier lepton shrinks the molecular bond to $\sim 500 \text{ fm}$, worth an effective $\sim 2.7 \text{ keV}$, and produces measured fusion at $10^8\text{--}10^{12} \text{ s}^{-1}$; the effective-energy formalism is thereby validated across seventy-eight orders of magnitude of rate. Contested and unreplicated: coherent correlated states [40, 41], whose claimed transparency gains of $10^{30}\text{--}10^{100}$ would dwarf every other rung — and which fail three further checks on this platform, beyond the drive-side closure of §5.4: the parametric drive is 8–75 $\times$ sub-resonant to the B–H stretch with sub-percent modulation depth, the formation time exceeds the plasmon coherence time by two orders, and the claims have no independent experimental confirmation. A NASA demonstration of beam-driven screened fusion in metal hydrides [52] and a 2026 muonic $p\text{--}^{11}\text{B}$ analysis [53] bracket what honest enhancement physics currently delivers: useful factors at keV beam energies, nothing at lattice energies.

Quantitatively: granting $U_{\text{e}} = 300 \text{ eV}$ — above the adiabatic bound — as an effective collision energy would set the $p\text{--}B$ per-collision penetration near 10^{-119} ; the contested 800 eV would give 10^{-73} . Both are beam-formula numbers (valid only for $U_{\text{e}} \ll E$, which fails here) applied outside their regime; for the bound molecular state the honest statement is that conduction-electron screening beyond the bond electrons already included shortens the barrier by at most tens of decades — a huge lever by any normal standard, and nowhere near the ~ 120 of §8.

Finding (screening). One mechanism, two signs: static screening of the drive ($\epsilon \approx 50\text{--}500$) raises the effective C1a threshold field by up to two and a half orders, while screening of the internuclear barrier is worth tens of decades on C3 at its measured magnitudes — a huge lever by any normal standard, against a gap of $\sim 120\text{--}145$ (§8), and with its largest claimed values contested. No screening of any kind touches the free-channel energy deficit.

8. Factor V — The Coulomb Barrier and the Strong-Channel Hierarchy (C3)

The binding route is the strict lattice analog of the solar pep reaction: a proton, an electron, and a host nucleus convert to the host's neutron-richer isobar plus a neutrino, with the neutron born bound. On the platform's majority host the channel is $e^- + p + {}^{10}\text{B} \rightarrow {}^{11}\text{B} + \nu_e$, exothermic by 10.672 MeV (Table 2). No drive is required; the channel is open at zero field, which is what makes it logically prior to the field route: bounding the free-neutron channel does not bound electron capture as a whole, because Eq. (2) is the loophole — and closing it honestly is this section's job.

What the capture populates matters as much as the Q-value. The neutrino phase space ($\propto E_\nu^2$) and the weak selection rules favor the strongly bound final states: the dominant outcomes are the ${}^{11}\text{B}$ ground state — a bare ~ 6 keV nuclear recoil, invisible to every detector in §13 — or bound excited states, which de-excite by 2–11 MeV gamma cascades (lines at 2.12, 4.44, 5.02 MeV). The alpha exit that reproduces this paper's triple signature opens only for capture into ${}^{11}\text{B}^*$ above 8.664 MeV (α to ${}^7\text{Li}$ ground state) and above 9.142 MeV for the 478 keV branch (α to ${}^7\text{Li}^*$), and the first states above that threshold face centrifugal suppression of their alpha widths. Without a calculated Gamow–Teller (spin-flip weak-transition) strength distribution — flagged in §15 as the first missing calculation — the honest statement is: the triple signature is a sub-branch, plausibly at the percent level or below, and the binding route's generic signatures are instead the gamma cascade and the silent ground-state channel. Section 12 builds the discrimination matrix on this basis.

The rate has the pep structure — weak vertex times electron contact density times nuclear overlap [20]:

$$\lambda_{\text{EC}} = [G_F^2 \cos^2 \theta_C (1 + 3g_A^2)/(2\pi^2)] \cdot |\psi_e(0)|^2 \cdot E_\nu^2 \cdot |M|^2 \cdot P_{\text{pB}}, \quad (11)$$

where P_{pB} is the probability of finding the bound proton at the boron nuclear surface and $|M|^2$ the (uncalculated) nuclear matrix element, set to unity for the purpose of this bound. With hydrogenic electron contact density the prefactor evaluates to $\approx 1.1 \times 10^{-7} \text{ s}^{-1}$ per site (Appendix D). Everything then hinges on P_{pB} — and here the choice of rate object is decisive. The free-collision Gamow factor at thermal or eV energies is the wrong object: its classical turning point at 1 eV lies 72 Å from the nucleus, sixty times outside the electron cloud that defines the actual potential. The proton is a bound, zero-point particle in a B–H bond of length 1.19 Å and stretch quantum $\hbar\omega_0 \approx 0.31 \text{ eV}$ (75 THz) [14]; the well-defined quantity is the zero-point tunneling probability through the screened molecular barrier — the formalism of Koonin and Nauenberg [54], whose D_2 -molecule benchmark ($\lambda_{\text{dd}} = 3 \times 10^{-64} \text{ s}^{-1}$, penetration $\sim 10^{-77}$) calibrates the method. Applied to the B–H geometry ($Z = 5$, reduced mass 0.92 u, chemical screening included), the penetration is 10^{-110} to 10^{-220} with central value $\sim 10^{-150}$, where the spread reflects the screening model — the dominant uncertainty. The capture rate is then

$$\lambda_{\text{EC}}({}^{10}\text{B}) \sim 10^{-160 \pm 50} \text{ s}^{-1} \text{ per B–H site,}$$

against a detection floor of $\sim 10^{-17} \text{ s}^{-1}$ per site (§13). The gap is $\sim 145 \pm 50$ orders of magnitude — vast, but a categorically different object from the field route's 10^8 : it is barrier arithmetic, not energy conservation, it is temperature-independent (zero-point, not thermal, tunneling), and every order of magnitude of genuine screening enhancement moves it. For the hydrogen overlayer the same formalism gives the platform's least-suppressed capture channel: the pep analog $p + e^- + p \rightarrow d + \nu_e$ between adjacent protons ($\approx 3 \text{ Å}$ spacing) at $\lambda \sim 10^{-80 \pm 15} \text{ s}^{-1}$ per pair — still ~ 63 orders below the floor, but closer to detectability than any survivable field-route drive by six full powers of ten in the exponent. Its product is cumulative deuterium, measurable by SIMS or HD mass spectrometry: a fourth, independent observable.

The binding route carries a corollary that restructures the whole proposal, and it must be stated bluntly. The pep reaction is the operative solar channel only because $p + p$ has no strong or electromagnetic exit. $p + {}^{10}\text{B}$ has both, wide open at zero energy: ${}^{10}\text{B}(p,\alpha){}^7\text{Be}$ ($Q = +1.146$ MeV, $S(0) \sim 10^3$ MeV·b through a near-threshold s-wave resonance [55, 56]) and ${}^{10}\text{B}(p,\gamma){}^{11}\text{C}$ ($Q = +8.690$ MeV); on the 80% isotope, ${}^{11}\text{B}(p,\alpha)2\alpha$ ($Q = +8.682$ MeV, $S(0) \approx 2 \times 10^2$ MeV·b [55]). These share the identical tunneling factor P_{pB} and beat the weak vertex by the ratio of S-factors — roughly 10^{22} – 10^{25} per contact. Three consequences. First, electron capture is never the first observable: any physics that enhances barrier penetration manifests as prompt ~ 0.7 MeV alphas and delayed ${}^7\text{Be}$ activity (478 keV at 10.4%, $t_{1/2} = 53.2$ d) some twenty-two decades before a single capture neutrino is emitted. Second, the experiment of §14 is therefore really a bound on the tunneling-enhancement product of the platform, with the strong channels as its live targets and the weak channel as an in-principle tail. Third, the founding motivation inverts: the binding route reinstates the very Coulomb barrier the neutron-first architecture [5] was designed to avoid, and is pursued here as honest bookkeeping, not as a rescue. The strong-channel rates under the same molecular overlap are $\lambda \sim 10^{-138 \pm 50}$ s⁻¹ per site — a gap of $\sim 120 \pm 50$ orders to the floor.

Finding (C3). The zero-point molecular overlap is $P \sim 10^{-150 \pm 50}$ — the central value already assuming aggressive screening (Appendix D) — and the weak channel pays a further $\sim 10^{-22}$ relative to its strong siblings, which therefore fire twenty-two decades earlier under any enhancement. C1b's satisfaction of the energy condition converts the proposal into barrier-limited, fusion-class physics: honest bookkeeping, not rescue, and the origin of every live target in the experiment.

9. Factor VI — The Electron at the Proton (C4)

Relativistic electron–proton corrections enter Eq. (11) only as modest multipliers, and it is worth itemizing them once to retire the question. The largest is Coulombic rather than relativistic: Sommerfeld focusing of slow electrons onto the proton formally enhances the contact density by $F = 2\pi\eta/(1 - e^{-2\pi\eta})$ with $\eta = \alpha c/v \approx 2.2$ for carriers at the Fermi velocity, giving $F \approx 14$. Three things demote F to an upper-bound scale rather than a multiplier. It is the same Coulomb physics as a bound 1s contact density — the rate may use the itinerant picture ($n_e \cdot F$, screened) or the bound picture ($|\psi_{1s}(0)|^2$), never both. Thomas–Fermi screening truncates the ~ 5 Å region over which the focusing builds (§7), cutting F toward order unity. And the proton's contact density is in any case dominated by its B–H bond electrons, for which a scattering formula is the wrong object — the DFT calculation of §15 is the right one. This bounded enhancement is what the upper end of Table 7's $|\psi_e(0)|^2$ row represents. Genuinely relativistic corrections are sub-percent: the contact-density exponent shifts by $(Z\alpha)^2$ -class factors — 0.005% at the proton, 0.13% at the boron nucleus — Darwin-term smearing of the contact interaction over the electron Compton wavelength is of the same order, the 10.67 MeV neutrino is already fully relativistic in the phase space of Eq. (11), and the proton is non-relativistic in every scenario this paper entertains (quiver ≤ 21 eV, §4). Relativity is decisive in this problem exactly once — in setting the field-route threshold $\gamma = 2.531$ (§3) — and decorative everywhere else.

The measured anchor deserves its own statement, because it is the factor's entire empirical content: environmental shifts of an actual electron-capture rate — ${}^7\text{Be}$ in C_{60} , in metals, at cryogenic temperature — are percent-level [24]. The bond's s-character sets the baseline contact density; gating moves the volume-normalized carrier density by parts in 10^3 . A 10 – 10^4 enhancement range is sometimes invoked for this factor; it has no identified physical source, and the defensible range is ~ 1 – 10 , pending the DFT calculation of §15.

Finding (C4). The electron is the least limiting factor: contact density at the proton is within $\sim 1\text{--}10\times$ hydrogenic under every defensible treatment, the largest available enhancement (Sommerfeld focusing, formally $\times 14$) is an upper bound already absorbed in that range, and genuinely relativistic corrections are sub-percent. Relativity is decisive exactly once in this problem — at the C1a threshold $\gamma = 2.531$ — and decorative everywhere else.

10. Factor VII — The Environment: Temperature, Pressure, and the Supercritical Unknown

The concept paper [1] specifies an operating environment this paper has so far priced only implicitly: the graphene/borophane fuel immersed in supercritical hydrogen at tens of megapascals and several hundred degrees Celsius (the chamber design point is 50 MPa and 873 K, rated to 80 MPa and 1073 K). Hydrogen's critical point is 33 K and 1.3 MPa, so the working fluid sits at $T/T_c \approx 26$ — a deep-supercritical dense gas ($n \approx 4 \times 10^{27} \text{ m}^{-3}$, $\rho \approx 14 \text{ kg}\cdot\text{m}^{-3}$ at 50 MPa), far from the critical point and therefore free of critical-opalescence fluctuation enhancement: “chaotic” here means dense-gas turbulence and thermal noise with Boltzmann statistics. Every material input this paper's bounds rest on — sustained-field ceilings and damage thresholds (§4), plasmon lifetimes (§6), screening models (§7), borophane hydrogen retention [14] — was measured or modeled at ambient or cryogenic conditions in vacuum, air, or inert atmospheres. To the author's knowledge, no published experiment and no quantitative model addresses the plasmonic or terahertz near-field response, the field-damage thresholds, or the dielectric screening of graphene — nor the stability and B–H chemistry of borophane — immersed in supercritical hydrogen, or in any dense supercritical fluid, at these conditions. The nearest literature treats each axis in isolation and in other regimes: finite-temperature plasmon theory and transient hot-carrier terahertz experiments assume vacuum or solid surroundings with the lattice near 300 K [57, 58]; damage thresholds are measured in vacuum or ambient air (§4); supercritical-fluid graphene work is exfoliation and synthesis chemistry with no in-situ electromagnetic characterization; and supercritical-hydrogen-on-carbon data are cryogenic adsorption isotherms. Worse for the proposal, the adjacent chemistry argues the environment is aggressive: hot molecular hydrogen etches graphene at ~ 0.1 MPa and 700–1000 °C [59], gasifies graphite to methane above ~ 800 °C at 5–10 MPa, and hydrogenates graphene toward graphane at gigapascal pressures [60]; borophane's thermal dehydrogenation window opens near 393 K at ambient pressure [61] — 873 K sits inside it — and whether 50–100 MPa of H_2 fugacity shifts that equilibrium back is, to the author's knowledge, unstudied in either experiment or theory. The chemical integrity of both sheets at the design point is therefore itself an open experimental question, stated as such rather than assumed. This is the paper's largest unmodeled materials uncertainty.

The regime changes the physics in identified ways, each currently unquantified. The bath itself is a broadband terahertz agitator: the wall-collision flux at 50 MPa is $3 \times 10^{30} \text{ m}^{-2}\text{s}^{-1}$ — about $1.3 \times 10^{11} \text{ s}^{-1}$ per B–H site — with ~ 70 fs collisions whose spectral content reaches the low-terahertz band, overlapping the plasmon dispersion though remaining thirty-fold below the 75 THz B–H stretch — broadband agitation that, by fluctuation–dissipation, can only thermalize the plasmon mode and add damping, never pump it coherently (Appendix E.1). Continuous hydrogen resupply can re-passivate desorption damage, plausibly raising the effective ceiling for repeated pulsing. Supercritical convection cools the sheet orders of magnitude faster than vacuum radiation, raising the sustainable dissipation — while, in the opposite direction, the dense neutral bath (rigorously plasma-free: Saha equilibrium leaves an ionized fraction of $\sim 10^{-46}$, not one free electron per laboratory volume) adds collisional plasmon damping that shortens τ .

These competing, unmeasured corrections motivate both the feedback analysis of Appendix E — which establishes what the regime cannot do regardless of the unknowns — and the in-situ ceiling measurement that §14 adopts as its zeroth deliverable.

Finding (environment). Thermodynamics caps what chaos can do: an isothermal bath cannot be converted into coherent field energy — zero Carnot efficiency, spontaneous-fluctuation probability $\sim \exp(-3.7 \times 10^6)$ for even a $(10 \text{ nm})^3$ mode — so the supercritical regime cannot rescue C1a regardless of its unknowns (Appendix E). Temperature likewise leaves C3 untouched: the 873 K Gamow peak sits 400 orders below zero-point tunneling, and vibrational excitation is worth a factor $\lesssim 3$ (Appendix E.5). The regime’s genuine unknowns are material — chemical survival of both sheets, damage ceilings, lifetimes — and all of them are measurable, which is why the experiment measures them first (§14).

11. Synthesis — The Channel Atlas

The factors now reassemble. Every channel available to the platform is a product of the factors bounded in §3–§10 — an energetic condition, times a delivery or overlap probability, times an electron factor, integrated under the survival and coherence constraints — and every claimed enhancement mechanism in the condensed-matter nuclear literature attacks exactly one factor, so its maximum effect can be priced on the channel it feeds. One number summarizes the stack: the unitary ceiling of the full binding-route enhancement stack — maximal screening, well-depth-limited squeezing, zero decoherence — still leaves every channel at least ten orders below the detection floor. That is the honest sense in which this paper “enhances the possibility” of low-energy electron capture: the possibility is real ($Q > 0$, an established reaction class), its least-suppressed incarnations are identified and ranked below, the levers are named with their measured sizes — and the remaining gap is stated, not hidden.

Table 4 assembles every channel discussed into one commensurable format: $\log_{10}$ of a rate in s^{-1} per named entity, with the prefactor, suppression exponent, physical conditions, and accessibility stated per row, and contested-mechanism rows segregated from computed physics. Rows above the detection floor would be observable; nothing is. The table is the paper’s honest center: it shows that the platform’s channels span ten thousand orders of magnitude of suppression, that the ordering inverts the field’s thirty-year emphasis — the weak free-neutron route is the most suppressed object in the table, the screened strong channels the least — and that the only rows within reach of even the most generous published enhancement claims are the d–d calibration channel and, far behind it, the pep analog.

Table 4. The channel atlas. Rate object: $\log_{10} \lambda$ in s^{-1} per the stated entity, at the stated conditions. “Penetration” rows give the barrier factor entering λ ; total λ includes the prefactor. Conventions and calibration in Appendix D; the detection floor is the HPGe row of Table 6 at $2.5 \times 10^{15} \text{ sites} \cdot \text{cm}^{-2}$. Field-route rows are suppression exponents multiplying a $\sim 10^{-7} \text{ s}^{-1}$ weak prefactor; their absolute rates are indistinguishable from zero at this table’s resolution.

Channel (per entity)	$\log_{10} \lambda \text{ (s}^{-1}\text{)}$	Dominant factor	Conditions	Status
Detection floor (per B–H site)	–17	HPGe 478 keV row, Table 6	$2.5 \times 10^{15} \text{ sites} \cdot \text{cm}^{-2}$	measured floor
d + d in interfacial D ₂ (per molecule)	–64	molecular tunneling, KN benchmark [54]	zero drive; calibration arm	established calc.

$p + e^- + p \rightarrow d + \nu_e$, H overlayer (per pair)	-80 ± 15	$Z = 1$ tunneling $\times$ weak vertex	$\approx 3 \text{ \AA}$ spacing; zero drive	estimate (App. D)
$p + {}^{10}\text{B} \rightarrow {}^7\text{Be} + \alpha / p + {}^{11}\text{B} \rightarrow 3\alpha$ (per site)	-138 ± 50	$Z = 5$ molecular tunneling	B–H bond, zero-point; screened	established calc.
$e^- + p + {}^{10}\text{B} \rightarrow {}^{11}\text{B} + \nu_e$ (per site)	-160 ± 50	same tunneling $\times$ weak vertex	as above	established calc.
Field route, quasistatic, survivable drive	$\lesssim -2 \times 10^8$	Zener exponent, Eq. (10)	$E \leq 10^{10} \text{ V} \cdot \text{m}^{-1}$	bound (§5.5)
Field route, quasistatic, hypothetical Van Hove drive	$\approx -6.3 \times 10^4$	Zener exponent	$3.3 \times 10^{13} \text{ V} \cdot \text{m}^{-1}$: lattice-destroying	bound (§5.5)
Field route, multiphoton, THz at threshold	$\approx -1.2 \times 10^9$	$ J_N(\xi) ^2$, Eq. (5)	$\xi = 2.32$: lattice-destroying	bound (§5.2)
CCS-enhanced variants of any row	(row) + 30...100 claimed	correlated-state transparency [40]	requires parametric drive at $2\omega_0$	contested, unreplicated

Reading the atlas downward: the d–d calibration channel sits 47 orders below the floor; the pep analog 63; the strong p–B channels ~ 120 ; bound-state capture ~ 145 ; and every field-route row lies deeper still — the least-suppressed, lattice-destroying hypothetical by a further two and a half powers of ten in the exponent, and any drive the material survives by six. If the contested CCS claims were right at their published magnitudes, the deuterated calibration arm is where this platform would light up — decades before any boron channel. That is precisely what makes it the right calibration arm, and what makes a null there meaningful for every row above it.

12. Signatures and the Discrimination Matrix

Whatever the rates, each channel family has a distinct fate, and the experiment’s power comes from reading them against each other. The neutron-mediated chain retains its clean template: an ultra-low-momentum neutron is captured by ${}^{10}\text{B}$,

$$n + {}^{10}\text{B} \rightarrow {}^7\text{Li} + {}^4\text{He} \quad (Q = 2.79 \text{ MeV}), \quad (12)$$

proceeding to ${}^7\text{Li}^*$ — and thence a 478 keV gamma — in 93.7% of thermal captures and to the ground state in 6.3% [2], with a 1.47 MeV alpha in the dominant branch and a thermal cross-section of 3837 b. (Natural boron contains 19.9% ${}^{10}\text{B}$; isotopic enrichment raises the capture fraction accordingly.) The energy emerges as charged particles plus the soft gamma — no hard primary. The other families differ in exactly the ways a coincidence experiment can read:

- **Bound-state capture on ${}^{10}\text{B}$ (binding route, weak).** Dominantly a γ cascade (2.12, 4.44, 5.02 MeV lines) or a silent ground-state transition with the neutrino removing up to 10.67 MeV; the heat-per-event is variable and below Q ; the 478 keV $+\alpha$ branch is a small sub-channel (§8). Requires ${}^{10}\text{B}$; absent in ${}^{11}\text{B}$ -enriched samples except via the ${}^{12}\text{B}$ channel.
- **Bound-state capture on ${}^{11}\text{B}$.** Makes ${}^{12}\text{B}$; β^- , $t_{1/2} = 20.2 \text{ ms}$, 13.4 MeV endpoint — a millisecond-lived activation signature, ideal for pulsed-drive lock-in counting in drive-off windows, and unique to the ${}^{11}\text{B}$ host.

- **Strong $p + {}^{10}\text{B}$ fusion.** Prompt ~ 0.73 MeV alphas plus ${}^7\text{Be}$ accumulation: a delayed 478 keV line growing with a 53.2-day half-life (10.4% of ${}^7\text{Be}$ EC decays [2]) that persists after the drive stops — cleanly distinguished from the prompt 478 keV of neutron capture. The (p,γ) sub-channel makes ${}^{11}\text{C}$: β^+ , 20.4 min, 511 keV annihilation — which sits 33 keV above the 478 keV line and dictates the HPGe resolution and a 511-coincidence veto.
- **Strong $p + {}^{11}\text{B}$ fusion.** Three alphas, continuum up to ≈ 5.7 MeV, ~ 8.7 MeV of charged-particle heat, no 478 keV at all — the aneutronic template, isotope-tagged.
- **pep analog in the overlayer.** Cumulative deuterium (SIMS / HD mass spectrometry); no prompt radiation accessible to this suite.
- **$d + d$ (deuterated calibration arm).** The measured pair: 2.45 MeV neutrons + ${}^3\text{He}$, and 3.02 MeV protons + t , in equal branches — detectable by the same alpha/charged-particle spectrometer and a neutron counter, with known bare cross-sections and the metal-screening literature [46–50] as the comparison baseline.

Absent X-rays. The objection that metallic-hydride claims lack the radiation their energetics imply is resolved differently per channel, and the resolution is now diagnostic rather than apologetic. The neutron-capture chain emits no hard primary — its 478 keV line is soft and specific. The binding route weak capture, by contrast, should emit 2–11 MeV cascade gammas in its dominant branches (§8): for that channel, hard gammas would be signal, and their absence in any claimed positive becomes evidence against it. The discrimination matrix (Table 5) turns the old embarrassment into a column.

Table 5. The discrimination matrix. Every claimed-signal scenario must reproduce its column pattern; ambient-neutron capture (first row) is the irreducible background (see “Ambient neutrons” below).

Channel	Prompt 478 keV	Delayed 478 keV (53 d)	α spectrum	β activity	Isotope / loading dependence
Ambient $n + {}^{10}\text{B}$ (background)	yes	no	1.47 MeV	no	needs ${}^{10}\text{B}$; tracks ambient flux, not drive
ULM $n + {}^{10}\text{B}$ (any n-first mechanism)	yes	no	1.47 MeV	no	needs ${}^{10}\text{B}$; tracks drive
EC on ${}^{10}\text{B}$ (binding route, weak)	weak ($\lesssim\%$)	no	weak, >8.7 MeV states only	no	needs ${}^{10}\text{B}$; γ cascade 2–11 MeV instead
EC on ${}^{11}\text{B}$ ($\rightarrow {}^{12}\text{B}$)	no	no	no	β^- , 20.2 ms	needs ${}^{11}\text{B}$; lock-in counting
$p + {}^{10}\text{B} \rightarrow {}^7\text{Be} + \alpha$ (strong)	no	yes, grows then decays	~ 0.73 MeV	${}^{11}\text{C}$: β^+ , 511 keV, 20 min	needs ${}^{10}\text{B}$
$p + {}^{11}\text{B} \rightarrow 3\alpha$ (strong)	no	no	continuum ≤ 5.7 MeV	no	needs ${}^{11}\text{B}$
$d + d$ (calibration)	no	no	$p(3.0 \text{ MeV})/t({}^3\text{He}) + 2.45 \text{ MeV } n$	no	needs D loading

Ambient neutrons. Owned explicitly rather than dismissed: ambient and cosmogenic thermal neutrons captured by the sample’s own ^{10}B produce the triple signature exactly, in exactly the predicted ratio — that is what makes the signature neutron-specific. The triple coincidence discriminates against instrumental artifacts; only drive correlation, isotopic nulls, and continuous flux monitoring discriminate against real ambient neutrons (§14). Likewise, atmospheric ^7Be — a ubiquitous cosmogenic surface contaminant — mimics the delayed 478 keV line and requires sealed blanks and pre-run assay.

13. Detection Floors

Because every channel is bounded (§11), we do not claim a predicted signal that beats noise. Instead we record the minimum surface activity each detector could see, so that the §14 experiment is framed as either a detection or a quantitative limit. Converting noise floors to an equivalent surface power density — at the stated convention of 2.79 MeV deposited per event, the $n + ^{10}\text{B}$ value — gives Table 6.

Table 6. Detection floors. The nuclear-product detectors are six to eleven orders of magnitude more sensitive than calorimetry. HPGe row: 10^{-3} cps noise in the 478 keV window, 5% absolute efficiency, 93.7% branch, steady state, 2.79 MeV per event; at 2.5×10^{15} B–H sites·cm $^{-2}$ this floor corresponds to 8.5×10^{-18} events·s $^{-1}$ per site — the reference line of Table 4. The last two rows are event-counting channels whose floors are set by track and afterpulse backgrounds; they are quoted as classes, to be quantified in protocol design. A calorimetric signal without accompanying nuclear products would be evidence against every channel in Table 5, not for one.

Detector	Noise floor	Min. detectable surface power
Isothermal calorimeter	~1 mW on 1 cm 2	~ 10^{-3} W·cm $^{-2}$
Quadrupole mass-spec (^4He)	~ 10^9 atoms·cm $^{-2}$ /week	~ 7×10^{-10} W·cm $^{-2}$
HPGe γ -spec (478 keV, prompt or delayed)	~ 10^{-3} cps, 5% eff.	~ 1×10^{-14} W·cm $^{-2}$
Si / CR-39 α spectrometry	track-background limited	~ 10^{-17} – 10^{-18} W·cm $^{-2}$ class
Pulsed-drive β lock-in (^{12}B , 20 ms)	afterpulse limited	~ 10^{-17} W·cm $^{-2}$ class

The quantitative inputs that would set the rates are listed in Table 7, with honest ranges. The surface field dominates the field route; the screening model dominates the binding route; both motivate the §14 scans.

Table 7. Parameter ranges. The $|\psi_e(0)|^2$ range is ~1–10, not the 10 – 10^4 sometimes invoked, which has no identified physical source (§9); the proton areal density (2.4 – 2.6×10^{15} cm $^{-2}$) reflects stoichiometric borophane.

Input	Plausible range	Status / note
Sustained surface field E_{rms}	$10^7 - 3 \times 10^9$ V·m $^{-1}$	damage-limited ceiling (§4); 10^{10} – 10^{11} values are tip-peak or single-cycle, not mode-filling; the dominant field-route variable
Static screening $\epsilon(q)$ at $q \approx 10^7$ m $^{-1}$	50 – 500	κ - and doping-dependent (§7); strong-field case open (§15)
Plasmon lifetime τ	0.1 – 1 ps	≤ 3 ps theoretical ceiling on hBN; coherence over N cycles not met by 10^{7-8} (§6)
$ \psi_e(0) ^2$ at the proton, vs hydrogenic	~1 – 10	anchored by percent-level ^7Be EC shifts [24]; DFT needed (§15)

Proton (B–H) areal density	$2.4 - 2.6 \times 10^{15} \text{ cm}^{-2}$	stoichiometric borophane [14]; B–H bond 1.19 Å, stretch 310 meV (75 THz)
Proton quiver energy U_p at 2.4 THz	2 meV – 21 eV	for $E_{\text{rms}} = 10^8 - 10^{10} \text{ V} \cdot \text{m}^{-1}$ (§4); keV scale inaccessible

14. The Experiment: A Calibrated Bound, Not a Decisive Test

No experiment here can be “decisive” about the mechanism in the usual sense: the dressed-mass threshold sits above the survivable window, and the §5 bound puts the modeled field-route rate tens of millions of orders below any floor — an experiment cannot sharpen that with six detectors and a season of beam time. What an experiment can do, uniquely, is bound the things theory does not fix: the tunneling-enhancement product of a driven two-dimensional hydride (the contested rungs of §7), and the existence of any unmodeled channel. Both targets are mechanism-agnostic; neither requires believing a single number in this paper. One sample family, six readouts, five arms — each arm a lever on a named condition of §2:

- **Field-amplitude arm.** Vary the THz plasmon drive across $10^8 - 3 \times 10^9 \text{ V} \cdot \text{m}^{-1}$ — the survivable window (§4), calibrated in situ by scattering-type near-field optical microscopy (s-SNOM) — and look for any super-linear turn-on in any channel of Table 5. Drive correlation is the primary discriminator against ambient-neutron background.
- **Frequency arm.** Add a mid-infrared pump at the B–H stretch ($\approx 2480 \text{ cm}^{-1}$, 75 THz) and at its parametric double ($\approx 150 \text{ THz}$, $\lambda \approx 2 \text{ } \mu\text{m}$): this is the modulation the CCS mechanism actually requires (§5.4, §7), which no THz drive supplies. A null here bounds the driven-correlated-state claim class on its own terms; the THz arm bounds the sub-resonant version.
- **Materials matrix.** Isotope swap (^{10}B -enriched / ^{11}B -enriched / natural), loading swap (H / D), substrate (suspended, hBN, SiO_2), temperature (4, 77, 300 K). The D-loaded samples are the calibration arm: d–d is the least-suppressed channel on the platform (Table 4), its bare cross-sections and metal-host screening literature are measured [46–50], and any genuine enhancement mechanism shows there first and largest. The isotope columns of Table 5 then identify any channel that does appear.
- **Supercritical arm.** Repeat the amplitude scan with the sample immersed in supercritical H_2 at the [1] design point (50 MPa, 873 K), behind a THz-transparent high-pressure window. Deliverable zero is materials, not nuclear: the in-situ sustained-field ceiling, plasmon lifetime, and damage threshold in the supercritical environment — quantities for which no published data or models exist (§10) and which set how tightly every other arm’s null constrains the contested mechanisms. Appendix E’s fixed-point model is the prediction this arm tests: continuous-wave fields saturating near $10^7 - 10^8 \text{ V} \cdot \text{m}^{-1}$, pulsed ceilings within an order of magnitude of the ambient damage limit, and no drive-correlated nuclear signal above the ambient-neutron baseline.
- **Readout and background suite.** Time-correlated calorimetry, ^4He QMS, HPGe gamma spectroscopy (prompt 478 keV; delayed 478 keV tracked over weeks for the ^7Be half-life; cascade lines 2.12/4.44/5.02 MeV; 511 keV coincidence veto for $^{11}\text{C}/^{13}\text{N}$ positrons, which sit 33 keV above the signal line), Si or CR-39 alpha spectrometry (1.47 MeV capture line vs 0.73 MeV ^7Be line vs $\leq 5.7 \text{ MeV}$ 3α continuum), pulsed-drive β lock-in for ^{12}B (20.2 ms) in drive-off windows, and post-run SIMS deuterium assay (pep channel). Continuous ^3He ambient-neutron monitoring; sealed-blank and

pre-run ^7Be assay; radon control. Null samples: borophane-free graphene, graphene-free borophane, and drive-off runs must all register no excess.

Cautionary precedent. The community’s recent history with environment-modified nuclear rates is instructive: the 2018 report of nuclear excitation by electron capture (NEEC) in $^{93\text{m}}\text{Mo}$ [62] exceeded theory by nine orders of magnitude and was subsequently excluded by a direct isomer-beam experiment [63]. A claimed condensed-matter weak-rate enhancement of many orders should be presumed wrong until it survives exactly the kind of background-resistant, multi-observable protocol that §14 specifies.

Outcomes. A positive result is read against Table 5: each channel family has a distinct column pattern, and no instrumental artifact produces a matched pattern across independent detectors. Any positive at the predicted ratios would already implicate physics outside both bounds of this paper — that is what makes it reportable. A null result is not empty: it places the first experimental ceiling on the tunneling-enhancement product of a driven 2D hydride — most stringently in the d–d calibration arm, where the gap between established theory and the floor is only ~ 47 orders (Table 4), so a null bounds any claimed enhancement mechanism to below that, on a platform purpose-built to maximize it. Either way the experiment measures something no theory in this paper fixes.

15. What This Paper Does and Does Not Claim

It does not claim that low-energy electron capture has been observed, nor that any channel’s rate is non-negligible — §5 and §8 establish the opposite, as bounds with stated assumptions and uncertainties. It claims four things. First, that the standard energy-gap objection is incomplete as usually stated: the 0.782 MeV deficit binds only the free-neutron channel, and the bound-final-state family (Table 2) is exothermic — established reaction physics (pep, bound-state β decay) at solid-state densities, suppressed by barrier penetration rather than forbidden by energetics. Second, that the field route’s coherence rescue fails for specific, explicit reasons — the n^N bonus cancels against g_0^N and the factorial energy denominators supply the residual hyperexponential suppression (§5.2, Appendix B) — and that the failure is robust across drive statistics (coherent, thermal, squeezed; §5.4) and across the Keldysh crossover (§5.5). Relative to the existing critiques [6, 7, 9, 10, 11], the contribution is to close the collective-coherence loophole quantitatively — the rebuttals’ promised many-body treatment [8] never appeared — and to do so in a form (Table 3) that is independent of whose field estimate one accepts. Third, that the least-suppressed low-energy capture channels are not where this field has been looking: they are the pep analog in the hydrogen overlayer and bound-state capture on the host nuclei, both identified and bounded here for what appears to be the first time in a condensed-matter setting, with the strong-channel hierarchy (§8) stated rather than hidden. Fourth, that a single calibrated experiment (§14) bounds the enhancement claims of this literature on their own terms, with the discrimination matrix (Table 5) separating six channel families and the irreducible backgrounds owned in advance.

Six calculations or measurements would close the remaining theory gaps, in priority order: (1) the Gamow–Teller strength distribution for $e^- + p + {}^{10}\text{B} \rightarrow {}^{11}\text{B}^*$, which sets the binding-route branching and the 478 keV sub-branch (§8); (2) a DFT calculation of the screened B–H molecular potential, the proton wavefunction at the boron nuclear surface, and the electron contact density at the proton, together with the strong-field screening of the drive at a charged adsorbate (§7, Appendix C) — replacing the ± 50 -order screening spread that dominates every binding-route exponent; (3) a coupling model for the THz plasmon and the 75 THz B–H stretch, without which the drive has no predicted effect on the binding route at all; (4) the strong-channel S-factor folding under the same molecular overlap, giving the absolute rates the §14 null

will bound; (5) a quantitative background budget (ambient and cosmogenic neutron flux, atmospheric ^7Be deposition, radon) for the §14 site; and (6) the material response of graphene and borophane in supercritical hydrogen — damage thresholds, plasmon lifetimes, cooling coefficients — for which neither data nor theory exist (§10) and which the §14 supercritical arm measures directly. None of these changes what the experiment is built to measure; (2), (3), and (6) change how tightly its null constrains the contested mechanisms.

The posture throughout is consistent: the mechanisms are bounded by theory; the bounds are statistics-robust, channel-complete, and factored into the conditions that any future claim must address one at a time; and the open questions — all of them — are matters of measurement.

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Appendix A. Threshold and the Dressed Mass

The cycle-averaged Volkov dispersion gives $\tilde{m} = m_e \sqrt{1 + \xi^2}$. Requiring $\tilde{m}c^2 \geq (m_n - m_p)c^2 = 1.293 \text{ MeV}$ gives $\sqrt{1 + \xi^2} \geq 2.531$, hence $\xi_{\text{th}} = 2.325$.

Appendix B. The N-Quantum Coupling: Exact Bookkeeping and Drive Statistics

Corrected vertex. The effective N-quantum vertex of Eq. (6) is generated at N-th order in the single-quantum coupling $g_0(a^\dagger \sigma_- + \text{h.c.})$, through N–1 virtual intermediate states. The k-th intermediate state has absorbed k quanta and is detuned by $\Delta_k \approx k\hbar\omega$, so the denominator product is $\prod_k \Delta_k \approx (N-1)!(\hbar\omega)^{N-1}$ — it grows factorially, not geometrically. (A constant denominator $\Delta \sim \hbar\omega_0$ would instead yield $\Gamma \propto \xi^{2N}$ — an

enhancement with $\log_{10} \approx +5.6 \times 10^7$ at $\xi = 2.3$, not a suppression; see §5.2.) Substituting into $\Gamma \propto |g_{\text{eff}}|^2 \langle n^N \rangle$ with the coherent-state moment $\langle n^N \rangle = n^N$ gives

$$\Gamma \propto (g_0 \sqrt{n/\hbar\omega})^{2N} / [(N-1)!]^2 \approx \xi^{2N} / (N!)^2 \approx (2\pi N)^{-1} (e\xi/2N)^{2N}$$

by Stirling — reproducing the Bessel asymptotic of Eq. (5). (The full strong-field amplitude is a generalized two-argument Bessel function; the single-argument asymptotic is the controlling factor at $N \gg \xi$.) The n^N coherence bonus is exactly cancelled by the g_0^N coupling cost; the $(N!)^2$ of the energy denominators is what remains. At $\xi = 2.3$, $\log_{10}|J_N|^2 \approx -1.4 \times 10^6$ for $N = 1.4 \times 10^5$, and $\approx -1.2 \times 10^9$ for $N = 7.8 \times 10^7$.

Drive statistics. For a thermal state the normally-ordered moment is $\langle (a^\dagger)^N a^N \rangle = N! \cdot \bar{n}^N$ — the largest phase-insensitive Gaussian value — giving the residual suppression $\Gamma_{\text{th}} \propto N! (e\xi/2N)^{2N} \bar{n}^N / \bar{n}^N \approx (e\xi^2/4N)^N$: still hyperexponentially small for all $N > e\xi^2/4 \approx 3.7$. Quadrature-squeezed states reach moments of order $(2N-1)!! \cdot \bar{n}^N$, changing the residual to $\approx (e\xi^2/2N)^N$ — fatal for all $N > e\xi^2/2 \approx 7.3$. No Gaussian statistics rescues any drive in Table 3. Beyond Gaussian states, the coherent-correlated-state program [40, 41, 42] claims fluctuation growth $G = 1/\sqrt{1-r^2}$; delivering $E_C = 782$ keV from 10 meV quanta needs $G \approx 8 \times 10^7$, i.e. $1-|r| \approx 8 \times 10^{-17}$, against $G \sim 10^3$ – 10^4 entertained in that literature, formation times of tens of picoseconds against plasmon coherence of 0.1–1 ps, and an independent fragility analysis [43]. Kinematic floor: $a^N |n\rangle = \sqrt{[n!/(n-N)!]} |n-N\rangle$ vanishes for $n < N$, so any rescue formula returning $n^* < N$ is unphysical on its face (Figure 3); the floor invalidates the formula but the magnitude of the bound rests on the dynamical argument above.

Appendix C. The Local Field and Static Screening

Geometric factor. For the 2D eigenmode the matrix element factorizes into an evanescent-decay factor $F_{\text{height}} = e^{-q^d}$, a hydrogenic form factor $[1+(qa/2)^2]^{-2}$, and a Debye–Waller factor $e^{-q^2 \langle \delta r^2 \rangle / 2}$. With $q = 10^7 \text{ m}^{-1}$, $d = 1 \text{ \AA}$, $a = 0.5 \text{ \AA}$, and $\langle \delta r^2 \rangle$ set by the B–H zero-point amplitude ($\sim 0.07 \text{ \AA}$), the product is $\eta_{\text{loc}} \approx 0.999$, dominated entirely by e^{-q^d} ; an RPA-impurity factor is sometimes included; it is dimensionally ambiguous as usually written and is omitted here, with impurity screening absorbed into the static dielectric function below. Static screening is quoted separately and is not small: $\epsilon(q) \approx 1 + q_{\text{TF}}/q$ with $q_{\text{TF}} = 4\alpha_g k_F / \kappa$ gives $\epsilon \approx 50$ – 110 (hBN-encapsulated, $\kappa \approx 4$ – 5 , $E_F = 0.2$ – 0.4 eV) up to $\epsilon \approx 250$ – 500 (suspended, $\kappa \approx 1$). The strong-field, non-linear screening of a $\sim 10^{9-11} \text{ V} \cdot \text{m}^{-1}$ longitudinal field at a charged adsorbate is an open calculation (§15).

Appendix D. The Bound-Final-State Channel: Kinematics, Windows, and Rate Inputs

Kinematics. With AME2020 atomic masses [3] (which make the electron bookkeeping automatic), $Q(e^- + p + {}^A Z \rightarrow {}^{A+1}Z + \nu) = S_n({}^{A+1}Z) - 0.782 \text{ MeV}$: for ${}^{10}\text{B}$, $S_n({}^{11}\text{B}) = 11.454 \text{ MeV}$ gives $Q = +10.672 \text{ MeV}$; the other rows of Table 2 follow identically, and the pep value $+1.442 \text{ MeV}$ matches the canonical solar number — the cross-check that anchors the bookkeeping. Exit windows in ${}^{11}\text{B}$: α -decay to ${}^7\text{Li}$ opens at 8.664 MeV and to ${}^7\text{Li}^*$ (the 478 keV emitter) at 9.142 MeV ; $S_n = 11.454 \text{ MeV}$. The first states above the α_1 threshold (9.185 MeV , $7/2^+$; 9.274 MeV , $5/2^+$) emit α_1 at only 43 and 132 keV against combined Coulomb-plus-centrifugal barriers, so realistic 478 keV emitters lie near 9.9– 10.6 MeV and the triple-signature branch is plausibly at or below the percent level pending the Gamow–Teller calculation of §15. Capture to bound states (the phase-space-favored majority) yields the γ cascade or the silent ground-state channel.

Rate. Eq. (11) with hydrogenic contact density $|\psi_{1s}(0)|^2 = 1/(\pi a_0^3) = 2.15 \times 10^{30} \text{ m}^{-3}$, $E_v = 10.67 \text{ MeV}$, and unit nuclear matrix element gives the prefactor $\approx 1.1 \times 10^{-7} \text{ s}^{-1}$ per site; the suppression is the probability P_{pB} of finding the bound proton at the boron nuclear surface. The free-collision Gamow factor is the wrong object for a bound proton (its turning point at 1 eV is 72 Å); the right one is zero-point WKB through the screened molecular barrier, calibrated to the Koonin–Nauenberg D_2 benchmark ($\lambda_{dd} = 3 \times 10^{-64} \text{ s}^{-1}$, penetration $\sim 10^{-77}$) [54]. For the B–H geometry (bond 1.19 Å, $Z = 5$, reduced mass 0.92 u) this gives $P_{pB} = 10^{-110} - 10^{-220}$ (central $\sim 10^{-150}$); the spread is dominated by the screening model and is the ± 50 carried through §8 and §11. The central value corresponds to an aggressive screening model (a constant $\sim 190 \text{ eV}$ barrier reduction, or equivalently a $\sim 0.16 \text{ Å}$ Yukawa screening length); an unscreened barrier truncated at the bond length gives $\sim 10^{-336}$, and a bare Coulomb barrier $\sim 10^{-5300}$ — so the central value is the mechanism’s best case, conservative for every negative conclusion drawn from it. The strong channels multiply the same P_{pB} by the ratio of S-factors — $S(0) \sim 10^2 - 10^3 \text{ MeV} \cdot \text{b}$ for the (p,α) channels [55, 56] against the pep-class weak equivalent $\sim 10^{-22} \text{ MeV} \cdot \text{b}$ [20, 21] — hence the $10^{22} - 10^{25}$ per-contact hierarchy of §8. For the overlayer pep analog (adjacent B–H protons at $\approx 3 \text{ Å}$, $Z = 1$, reduced mass $\frac{1}{2}m_p$), the same calibrated WKB gives a per-pair penetration that combines with the weak prefactor to $\lambda \sim 10^{-80 \pm 15} \text{ s}^{-1}$ — the platform’s least-suppressed capture channel. Floor conversion: the HPGe row of Table 6 ($10^{-3} \text{ cps} / 0.05 \text{ efficiency} / 0.937 \text{ branch} = 2.1 \times 10^{-2} \text{ events} \cdot \text{s}^{-1} \cdot \text{cm}^{-2}$) at $2.5 \times 10^{15} \text{ sites} \cdot \text{cm}^{-2}$ is $8.5 \times 10^{-18} \text{ s}^{-1}$ per site, $\log_{10} = -17.1$: the reference line of Table 4.

Appendix E. Energy Feedback at the Driven Supercritical Interface

This appendix models the question the operating environment of [1] raises: can the chaotic conditions of a driven, supercritical-hydrogen interface — and the feedback between drive, heating, and material response — escalate until electron capture becomes possible? The answer separates cleanly into what energy conservation forbids regardless of the unknowns, and what the unmeasured material parameters of §10 leave open.

E.1 The bath. Supercritical H_2 at the design point (50 MPa, 873 K) is a dense neutral gas: number density $n = P/k_B T \approx 4.2 \times 10^{27} \text{ m}^{-3}$ ($\rho \approx 14 \text{ kg} \cdot \text{m}^{-3}$; twice these at 80–100 MPa), thermal scale $k_B T = 75 \text{ meV}$, internal energy density $u_{\text{bath}} \approx (5/2)nk_B T \approx 1.3 \times 10^8 \text{ J} \cdot \text{m}^{-3}$. It is deep-supercritical ($T/T_c \approx 26$), so its fluctuations are Boltzmann-Gaussian, not critical. It is rigorously plasma-free: the Boltzmann factor for hydrogen ionization is $\sim 10^{-79}$, and the full Saha equilibrium — through the $\sim 10^{-12}$ molecular-dissociation step — leaves an ionized fraction of $\sim 10^{-46}$, not one free electron in a laboratory volume; thermionic injection from graphene (work function 4.5 eV) is suppressed by $\sim 10^{-26}$. Its agitation of the interface is quantified by the wall flux, $3 \times 10^{30} \text{ m}^{-2} \text{ s}^{-1}$: each B–H site is struck $\sim 1.3 \times 10^{11}$ times per second by $\sim 70 \text{ fs}$ collisions whose spectral weight reaches the low-terahertz band. By fluctuation–dissipation this broadband noise can only thermalize the plasmon mode — toward $n_{\text{th}} \approx k_B T / \hbar \omega \approx 5 - 8$ quanta, microscopic fields — and add damping; it cannot pump coherently. For the 75 THz stretch the bath is an ordinary thermostat (first-vibrational occupation $e^{-\hbar \omega_0 / k_B T} \approx 1.6\%$).

E.2 The feedback loop and its fixed points. The loop runs: drive power $\rightarrow$ plasmon field $E \rightarrow$ dissipation of the stored areal energy $u_A \approx \frac{1}{2} \epsilon_0 E^2 d$ (mode height $d \approx 1 \text{ nm}$) each lifetime $\tau \approx 1 \text{ ps} \rightarrow$ local heating $\rightarrow$ removal by supercritical convection at q_{cool} . A continuous drive therefore saturates at the balance point $u_A / \tau = q_{\text{cool}}$, i.e. $E_{\text{cw}} = \sqrt{(2q_{\text{cool}} \tau / \epsilon_0 d)} \approx 1.5 - 5 \times 10^7 \text{ V} \cdot \text{m}^{-1}$ for $q_{\text{cool}} = 10^6 - 10^7 \text{ W} \cdot \text{m}^{-2}$ — generous values even for supercritical heat transfer. (Holding the field-route threshold field in a single mode would dissipate $\sim 0.1 \text{ W}$ per mode, $\sim 10^{14} \text{ W} \cdot \text{m}^{-2}$ through its footprint: eight orders beyond any cooling.) Pulsed operation evades

the steady balance up to the damage ceiling — ambient-measured at $\sim 3 \times 10^9 \text{ V} \cdot \text{m}^{-1}$ (§4), with an unknown supercritical correction that is precisely the zeroth deliverable of the §14 supercritical arm; the unknown plausibly points down, since at 873 K in chemically reactive hydrogen (§10) the ceiling should if anything be lower than its ambient value, which strengthens every bound that uses it. Feedback through heating, desorption, or carrier generation can move the system between these fixed points; it cannot move the fixed points themselves, which are set by τ , q_{cool} , and the damage physics.

E.3 The thermodynamic cap. Whatever the feedback mechanism, two ledgers bound it. Energy: reaching ξ_{th} requires a field energy density $\frac{1}{2}\epsilon_0 E_{\text{th}}^2 \approx 4.4 \times 10^{10} \text{ J} \cdot \text{m}^{-3}$ — roughly 350 times the entire internal energy density of the 50 MPa bath ($\approx 175\times$ at 100 MPa; real-gas corrections widen both ratios). Entropy: concentrating bath energy from a larger volume is not forbidden by arithmetic — a micron of adjacent fluid contains the areal requirement — but it is forbidden by the second law: the interface and bath are isothermal, so converting bath heat into coherent, low-entropy field energy proceeds with zero Carnot efficiency, and as a spontaneous fluctuation the threshold field in even a $(10 \text{ nm})^3$ mode volume has probability $\sim \exp(-\frac{1}{2}\epsilon_0 E_{\text{th}}^2 V / k_B T) \approx \exp(-3.7 \times 10^6)$ — a number that makes the field-route rate exponents look generous. The only reservoir that can do work on the mode is the external drive, and the drive is capped by E.2. The escalation scenario fails at the ledger, independently of every unmeasured material parameter: no chaotic intensification of an equilibrium bath can reach the dressed-mass threshold, even transiently.

E.4 Why chaos does not beat the budget. The capture exponents are hyperexponential in the field, so a fluctuating drive's time-averaged rate is dominated by its largest excursions, not its mean — formally, a Gaussian field-noise model puts the optimal fluctuation at hundreds of standard deviations. That is the correct lesson read backwards: when the saddle point sits hundreds of sigma out, the Gaussian tail is fiction, and the physical cutoff — the maximum field the local energy budget and the material survive — controls the answer. E.2 and E.3 supply that cutoff. Chaotic conditions reshuffle when the maximum occurs; they do not raise it.

E.5 Temperature and the binding route. The same discipline applied to the barrier-limited channels: at 873 K the free-collision Gamow peak for $p + {}^{10}\text{B}$ sits at $E_0 \approx 32 \text{ eV}$ with a penetration exponent of ≈ -550 in $\log_{10}$ — four hundred orders worse than the bound-state zero-point value (§8) — and the free channel would in any case need free protons, of which the neutral molecular bath has essentially none (atomic fraction $\sim 10^{-12}$, ions Saha-suppressed far below). Thermal collisions are irrelevant at any temperature the chamber survives; the zero-point molecular rate remains the honest binding-route number. Thermal excitation of the B–H stretch is likewise bounded: across honest screening models, the 1.6% first-vibrational population raises the penetration exponent by at most ~ 2 orders out of 150, a factor $\lesssim 3$ in total rate — inside the quoted ± 50 -decade spread. What the supercritical regime genuinely offers is material, not nuclear: a possibly different pulsed ceiling (unmeasured, plausibly lower; E.2), continuous hydrogen resupply that re-passivates the surface and extends integration time at maximum drive — set against the open question of whether the sheets chemically survive at all (§10) — and efficient heat removal. Every escalation pathway from chaotic conditions to electron capture must pass through one of the named material unknowns; each is measurable, which is why §14 measures them first.